Unit I — Logic and Proofs

Propositional Logic & Hypothetical Syllogism for Medical Triage

GROUP 4 • Discrete Mathematics • August 28, 2026

Educational mathematics assignment — not medical advice.

1. Case Title (Unit I Focus)

Proving that presenting both sentinel symptoms A and B justifies further testing T, via the chain $(A \land B) \to X \to T$, where X = "disease X suspected".

2. Problem Statement (Unit I)

Given premises P_1: $(A \land B) \to X$ and P_2: $X \to T$, does $(A \land B)$ alone logically entail T? The answer must be proved, not assumed, and cross-checked semantically.

- If $A \land B$ is true, can we derive X and then T by Modus Ponens?
- If $A \land B$ is false, does the implication hold vacuously?

3. Real-world Context

In triage chatbots, the rule "two red-flag symptoms together trigger a workup" must be auditable. Clinicians ask: does the rule follow from the knowledge base, or is it an ad-hoc shortcut? A proof provides the audit trail that a black-box score cannot.

4. Mathematical Concept Used

Propositional logic: propositions A, B, X, T; connectives $\land$, $\to$; inference rules Modus Ponens, Conditional Proof, and the derived Hypothetical Syllogism. Truth-table semantics with $2^4 = 16$ valuations as a decision procedure for validity.

5. Mathematical Formulation

Variables as above. Premises: $P_1 := (A \land B) \to X$, $P_2 := X \to T$. Theorem: $\vdash (A \land B) \to T$.

6. Analysis / Solution — Proof

Hypothetical Syllogism, presented as a Conditional Proof:

Step	Statement	Justification
1	$(A \land B) \to X$	Premise P_1
2	$X \to T$	Premise P_2
3	$A \land B$	Assumption (for CP)
4	X	Modus Ponens 1,3

5	T	Modus Ponens 2,4
6	(A ∧ B) → T	Conditional Proof 3–5 ∎

7. Diagram / Truth Table

Full 16-row table. Shaded rows: both premises true (1 fired + 9 vacuous). In every shaded row the conclusion is true — validity holds. Vacuous rows illustrate “no false alarm” when A∧B is false.

A	B	X	T	A∧B	(A∧B)→X	X→T	(A∧B)→T
T	T	T	T	T	T	T	T
T	T	T	F	T	T	F	F
T	T	F	T	T	F	T	T
T	T	F	F	T	F	T	F
T	F	T	T	F	T	T	T
T	F	T	F	F	T	F	T
T	F	F	T	F	T	T	T
T	F	F	F	F	T	T	T
F	T	T	T	F	T	T	T
F	T	T	F	F	T	F	T
F	T	F	T	F	T	T	T
F	T	F	F	F	T	T	T
F	F	T	T	F	T	T	T
F	F	T	F	F	T	F	T
F	F	F	T	F	T	T	T
F	F	F	F	F	T	T	T

Table 1 — Full truth table for Unit I. Shaded rows satisfy both premises.

8. Interpretation of Results

Validity means the system never recommends testing without a derivation when both premises are adopted. The 9 vacuous rows are not failures — they are the desired “no alarm” behaviour when the antecedent is false. The single fired row (T,T,T,T) is the critical positive pathway that the live Symptom Checker animates.

9. Practical Application

The proved chain is deployed as the Symptom Checker card: toggles for A/B drive the forward chain live, with an inference strip $(A \wedge B) \to X \to T$ lighting sequentially or a vacuous-case explainer when the rule does not fire.

10. Innovation / Proposed Improvement

- Add temporal logic (symptoms with onset times) and probabilistic thresholds (e.g., “X with $p > 0.7 \to T$”).
- Surface the trace (which premises fired) alongside every recommendation for clinician override.

11. Conclusion

The triage rule is sound, complete for its fragment, and explainable — a pragmatic foundation before any ML ranking is layered on.

12. References

- Rosen, Discrete Mathematics, 8th ed., McGraw-Hill, 2019.
- EU AI Act & FDA SaMD guidance — motivation for auditable logic.

All counts and proofs in this report are computationally verified (see live site https://ai-medical-diagnosis-site.vercel.app).